

CASSANDRA MCGINLEY

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[Cassandra McGinley Website](#)

EDUCATION

Embry-Riddle Aeronautical University (ERAU)

Daytona Beach, FL

Bachelor of Science in Astronomy and Astrophysics (*with Honors*), Minor in Applied Mathematics

May 2027

Honors: Sigma Pi Sigma (SPS), Dean's List

GPA: 4.00/4.00

Relevant Coursework:

Astronomy: Introductory Astronomy and Astrophysics I (Stars and Planetary Science) & II (Galaxies, ISM, & Cosmology); Introductory Astrophysics Lab; Stellar Astrophysics; Observational Astronomy (utilized ERAU 1-meter); Cosmology (In-Progress)

Advanced Physics: Modern Physics; Classical Mechanics; Thermodynamics and Statistical Mechanics; Applied Quantum Mechanics (In-Progress); Electricity and Magnetism (In-Progress)

RESEARCH EXPERIENCE

Coincidental Events and Signal Reflection/Refraction at RNO-G Stations

Astroparticle Physics Group, Department of Physics, University of Kansas

May 2026 – Present

Invited to continue as a remote research assistant during the academic year.

Project: Analyzing coincidental radio-signal events across multiple RNO-G stations & analyzing real and simulated refracted and direct waveforms at varying ice depths

Advisor: Dr. David Z. Besson

Results: Utilized cross-station events to identify coincidental signals found at more than 2 stations. Computed refracted and direct signal amplitude ratios and time delays to compare real calibration data to simulation.

Brightness & Structure Analysis of JWST Galaxies

Extragalactic Group, Department of Physical Sciences, ERAU

April 2025 - Present

Project: Model and analyze JWST Galaxy NGC 4406 for its 1-dimensional and 2-dimensional brightness profiles to compare its structure to previous HST literature

Advisor: Dr. Bililign Dullo

Results: Created both 1-dimensional and 2-dimensional brightness fits of NGC 4406, found best-fit structural parameters of NGC 4406, completed an analysis of multiple NGC galaxies

Connecting TESS Wide-Binary Stellar Activity with Rotation and Age Through Gyrochronology

ERAU Gyrochronology Team, Department of Physical Sciences, ERAU

May 2024 - Present

Project: Building a pipeline to investigate how stellar flare frequency and intensity of TESS Wide-binary stars correlates to Gyrochronology parameters, such as rotation period, color, and age.

Advisor: Dr. Terry D. Oswalt

Results: Built a pipeline to intake stellar IDs and pull lightcurves from MAST archive. This pipeline also computes stellar flare frequency and intensity. Used these values and previously known parameters to create a comprehensive 3-parameter plot that shows a decrease in flare intensity as rotation period gets larger (as the star ages).

Developing Diagnostic Systems on Pellets at Columbia (PAC) Test Stand

PAC Research Group, Columbia Plasma Physics Lab, Columbia University

June 2025 - May 2026

Invited to continue as a remote research assistant during the academic year.

Project: Operating and developing PAC fusion technology test stand diagnostic systems, such as camera, microwave cavities, and pellet tracking.

Advisor: Dr. Carlos Paz-Soldan

Results: Utilized microwave cavity system of PAC to save pellet signals to compute mass, individually operated and repaired PAC as needed, analyzed pellet deflection angles to prepare for e-beam ablation of pellets. Tested/repared vacuum leaks, hardware malfunctions, and electronic circuitry

Hands-on Observation of Messier 51's H α Content

ERAU 1-Meter Observatory, ERAU

January 2026 - May 2026

Project: Observing Messier 51 using the ERAU 1-meter telescope to study radial trends in H α emission.

Advisor: Dr. Ted von-Hippel

Results: Reduced and analyzed over 60 hours of imaging data to measure relative H α intensities of H II regions as a function of galactocentric radius. Successfully compared measured H α radial trends with published literature to evaluate consistency with previous studies

PUBLICATIONS AND MANUSCRIPTS

McGinley, C., Oswalt, T. D., Lares-Martiz, M., 2026, "Connecting Stellar Activity with Rotation through Gyrochronology" Research Notes of the American Astronomical Society, in preparation.

Dullo, B., Creviston, E., Stanley, M., Krauss, S., **McGinley, C.**, et al., 2026, "The Depleted Core of Early-type Galaxies with JWST," Astrophysical Journal, in preparation.

McGinley, C., Creviston, E., 2026, "Radial Distribution of H II Region Relative Intensities in the M51 System," Embry-Riddle Aquilla Observatory Journal, in preparation.

McGinley, C., Wyckoff, E., Desrochers, G., Creviston, E., 2026, "Twilight and Dark Sky Flat Calibration Analysis of ERAU 1-meter Telescope," Embry-Riddle Aquilla Observatory Journal, submitted.

POSTER PRESENTATIONS

APS-DPP Research Poster Presentation

November 2025

"Diagnostics and Operation of the Pellets at Columbia Test Stand," APS-DPP 67th Annual Meeting, Main Presenter and Author.

Florida Academy of Sciences Research Poster Presentation

March 2025

"Analyzing Relationships Between Stellar Parameters and Stellar Flare Activity," Florida Academy of Sciences 88th Annual Meeting, Main Presenter and Author.

AAS iPoster Presentation

January 2025

"Linking Stellar Parameters with Activity of Wide Binaries with TESS," 245th Meeting of the American Astronomical Society, Main Presenter and Author.

More posters and PDFs are detailed on my [website](#)

HONORS & ACHIEVEMENTS

Sigma Pi Sigma Honor Society

2026

For demonstrated excellence in physics. Awarded by Department of Physics at ERAU for completing all physics courses with a B letter grade or higher and maintaining over a 3.6 GPA.

ERAU "Shining Star" Employee Award

2025

Awarded to one student employee per ERAU employer who demonstrates leadership and dedication to their work.

Edward W. Stimpson Endowed Scholarship

2025

For outstanding community leadership and service.

Best Physical Sciences Poster

2025

Awarded Best Physical Sciences Poster at the 88th Annual Florida Academy of Sciences Meeting.

John G. and Isabel S. McKay Endowed Scholarship

2024

Award intended to recognize hardworking and devoted university students.

ERAU Women of Excellence Award

2023

Awarded to dedicated women in their field of study at ERAU.

EXTRACURRICULAR EXPERIENCE AND SERVICE

ERAU College of Arts and Sciences (COAS) Student Council

Daytona Beach, FL

Astronomy and Astrophysics (AA) Representative

May 2025 - Present

- Coordinates a peer-mentoring program for first-year students in the ERAU AA program
- Successfully advocated for Cosmology to be offered as an elective to rising fourth-year ERAU students

The Brown (Formerly MOAS)

Daytona Beach, FL

Planetarium Operator and STEM Outreach Volunteer

July 2024 - Present

- Volunteers at special events at the Brown Museum, such as for Space Day, Women and Girls in Astronomy Day, and many more.
- Prepared and presented 2 original shows, including *Rings and Realms: The Moons of Saturn* and *The Early Universe*

ERAU Astrophysics Research Club

Daytona Beach, FL

Secretary

August 2023 - Present

- Former Co-President and Vice-President in charge of coordinating meetings, special club events, and fundraisers
- Provides and connects new students (first and second-years) with introductory Astrophysics and Aerospace research opportunities at ERAU

ERAU Emergency Response Team

Daytona Beach, FL

Emergency Medical Technician (EMT)

August 2023 - January 2025

- Former Clearing Crew Chief responsible for medical care of patients during designated shifts
- Provided over 150 hours of free, unpaid, medical care to ERAU students and faculty

TECHNICAL SKILLS

Programming:	Python, Java, HTML
Analysis & Visualization:	Data analysis, matplotlib, statistical modeling
Hardware/Lab:	Vacuum systems, soldering, RF antenna measurement
Operating Systems & Tools:	Linux, Git, Jupyter Notebook, LaTeX
Other Software:	DS9, AstroImageJ